ON-SITE HOLOGRAPHIC BUILDING CONSTRUCTION

A Case Study of Aurora

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Abstract. Geometrically complex building components' reliance on high-touch implementation often results in tedious information reprocessing. Recent use of Mixed Reality (MR) in architectural practices, however, can reduce data translation and potentially increase design-to-build efficiency. This paper uses Aurora, a single-story residential building for 2021 China's Solar Decathlon Competition, as a demonstrator to evaluate the performance of on-site holographic building construction. This paper firstly reviews recent studies of MR in architectural design and practice. It then describes an MR-aided construction process of Aurora's non-standard building envelope and rooftop mounting structure, where in-situ holographic registration, human-machine cooperation, and as-built analysis are discussed. This paper concludes by stating that MR technologies provide unskilled implementers with a handy approach to materialise complex designs. The research was guided by the UN Sustainable Development Goals, especially aligning with the GOAL 9 which seeks innovations in industry and infrastructure.

Keywords. Mixed Reality; Non-standard Architecture; Low-tech Construction; Solar Decathlon Competition; SDG 9.

1. Introduction

Architects' use of computation-driven design techniques has widely expanded their design solution space, leading to a proactive exploration of complex architectural forms. Conventional design annotation methods may appear inefficient, especially in construction environments lack of advanced fabrication means and skilled labourers, the 3D-to-2D translation of non-standard design intent can be tedious and repetitive. The recent use of Mixed Reality (MR) among experimental architectural projects, however, suggests a new path for virtual-actual interactions. By projecting computationally generated geometries onto the field of vision with wearable devices or smartphones, it creates an immersive environment for implementers. The interactive holography allows architects to translate design intentions to intelligent instructions instead of static plans and sections, offers a significant opportunity to

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reduce construction complexity, and potentially increases precision (Jahn et al., 2018). MR-based architectural design practices are situated within a post-digital context, where the humanisation of digital is addressed via the interplay between digital and analogue culture, between high-tech and high-touch experiences, between global and local matters (Crolla, 2018). It presents a refined human-machine interaction that extends human intuition and tactile sense, and encourages collaborative knowledge that emerged from on-site serendipity. By constructing an information feedback loop between the as-built condition and the digital model, MR-based architectural practice improves designers' capacity to respond and adapt to local idiosyncrasies.



Figure 1. Aurora in snow (Zhang Jiakou, China, 2021)

This research aligns with the GOAL 9 of UN Sustainable Development Goals which seeks innovations in industry and infrastructure. It uses Aurora, a single-story passive residential building designed for 2021 China's Solar Decahedron Competition (SDC), as a demonstrator project to discuss the process and performance of on-site holographic construction (Figure 1). Aurora was built in Zhang Jiakou, China, in August 2021 with its main steel structure and partial components had been prefabricated in a Suzhou factory. In order to increase resource-use efficiency and reduce construction waste, the building was transported and assembled in modules. The challenge left to the construction team, consisting of mostly college students with no on-site experience, is the realisation of a non-standard building envelope made of flipped panels and a laminated bamboo mounting structure for the solar system. With the help of holographic instructions provided by Microsoft HoloLens and smart devices, unskilled students were able to assemble these parts in a short time with precision.

This research aims to illustrate human's extended implementation capacity enhanced by MR technology and whose feasibility for real-world architectural practices. It argues that such human-machine interaction breaks away from a conventional paradigm of design annotation, allows on-site implementers to directly communicate with digital design models, therefore stimulating a successful delivery of non-standard design.

2. MR in Building Practice

Paul Milgram and Fumio Kishino (1994) defined MR as a continuum of virtual reality technologies that overlays virtual objects with in-situ conditions. Today, MR is ubiquitously available to the general public with SDK such as ARCore and ARKit for smart devices. MR-based applications, therefore, have been adopted across various research fields including design, education, and entertainment. Within building industries, most fabrication and construction processes still heavily rely on human operations despite the dramatic rise of robotic automation. MR technologies, comparatively speaking, aim to augment manual operations instead of removing human factors off the equation. By providing holographic visual guidance like operating checklist and instructions, architectural designers can maximize the use of building information model.

Recent studies mainly focus on MR-aided architectural design innovations, human-centric manual fabrication and assembly tasks. Meža et al. (2015) applied holographic instruction to track and monitor the process of construction projects; Fazela and Izadi (2018) introduced an interactive tool for constructing complex modular surfaces in Mix Reality environment; Goepel (2019) tested MR with HoloLens in the assembly of non-standard prefabricated elements based on an optimized parametric structure. Mitterberger et al. (2020) referred to the major deficiency derived from the previous MR-aided manual constructions is the insufficient alignment of the digital model with the physical environment. In response, they presented a customized MR system based on an object-based visual-inertial tracking method in the construction of a brick envelope. This system, however, requires future development of app-based applications for HMD or smart devices otherwise cannot be pervasively feasible to building constructions across different contexts.

Maker-based 3D registration technique is currently widely used in Virtual Reality (VR) and MR applications. Developed by Gwyllim Jahn, Cameron Newnham, and Nicholas van den Berg, Fologram is among the popular MR applications adopted by architectural researchers, designers, and builders. Its capacity to connect with the parametric procedural modelling allows users to make ad-hoc design modifications according to site-specific conditions and restrictions. The dynamic holographic projection that Fologram creates can potentially establish a real-time feedback loop between the virtual objects and the actual builds. Research projects worked with Fologram, such as Timber De-Standardized (Lok et al., 2021), Augmented Grounds (Hahm et al., 2020), and Steampunk Pavilion (Jahn et al., 2019), revealed a strong bond between design intention, visual instructions, and tangible operations.

However, outdoor building-scale experiments still require further investigation because the accuracy of a marker-based registration can easily be influenced by matters such as lightning condition, project size, and user's field of vision. This paper discusses an MR-aided construction process of Aurora's non-standard envelope and mounting structure. It examines the potential use of Fologram and HoloLens in real-world building projects within an outdoor environment. The major research goals include the increase of marker-tracking accuracy by working with associative modelling, the test of system stability within volatile environment, and the exploration of in-situ human-machine collaboration.

2. The Aurora Case

As a joint research project carried out by Technical University of Denmark and Soochow University, Aurora's design and construction aimed to demonstrate the integration of energy-efficient systems, the use of environmental-friendly building materials, and the adoption of BIM for paperless project delivery. The entire realisation process lasted a month including off-site prefabrication and on-site assembly.

2.1. DESIGN OF BUILDING ENVELOPE

Aurora's envelope was designed to study how computation can offer design opportunities while maintaining construction rationality for the sake of a high-touch implementation. The design of the continuously varying facades not only intended to address an image of Aurora but also to respond to façade performance. With McNeel Rhinoceros and its procedural modeller plug-in Grasshopper, the pattern was visualized through differentiated flipping of bamboo panels which have been designed with the same height of 100mm but in different lengths due to openings of doors and windows. There are in total 2303 panels of 42 types used in Aurora's envelope and they are staggered in a vertical direction and overlaid by 20mm for the sake of water penetration (Figure 2).



Figure 2. The use of laminated bamboo material in Aurora's façade

The shape of the pattern was influenced by a NURBS control curve and the solar radiation of the site. Two major variables were determining the flipping angle of a bamboo panel: the relative distance between the geometric centre of the panel unit and its closest point on the control curve; also, the solar radiation intensity it gets during a summer day. The closer a panel to the curve and the more energy it receives; the larger angle it flips (Figure 3). These flipped panels help to reduce exterior wall temperature and humidity in summer by stimulating passive ventilation in between façade layers.

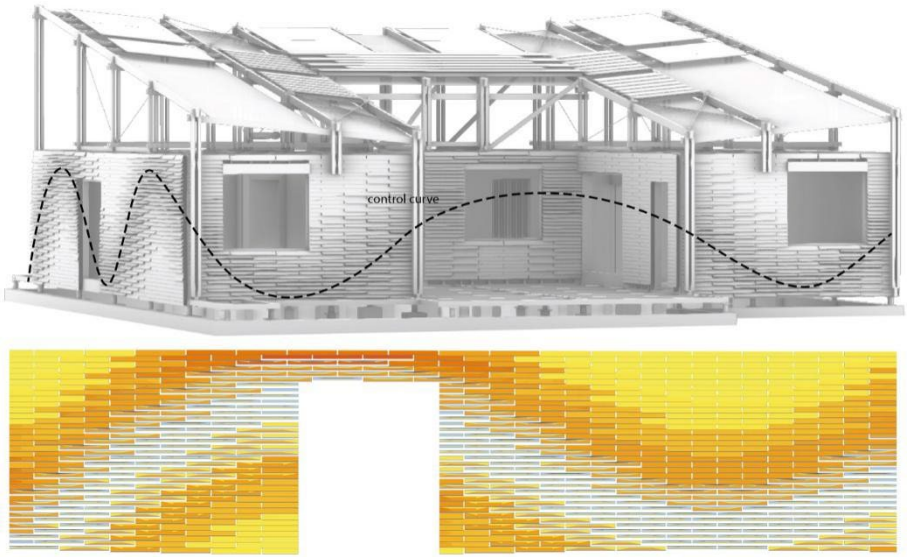


Figure 3. Design variables defining panels' flipping angles: a control curve (up) and the intensity of solar radiation (bottom)

2.2. DESIGN OPTIMIZATION AND PREFABRICATION

A visually smooth pattern is made of continuous differentiated flipping panels which, however, was impossible to realize in this case. Since the students decided to take over façade materialisation, the initial design intention must be optimized to accommodate a high-touch implementation process by unskilled labourers. Figure 4 shows an assembly diagram of the non-standard building envelope, where the bamboo panels were cut into needed lengths and mounted on vertical wood keels using self-tapping screws. The flipping mechanism is the angle blocks connecting the panels and the keels, and these wood blocks had to be manually fabricated by students and later installed in position. Hence, in order to seek a feasible design solution not only meets the aesthetic demand and the ventilation performance, but also accommodates low-tech operations, the design team eventually adopted four types of flipping angles in 25, 40, 55, and 70 degrees.

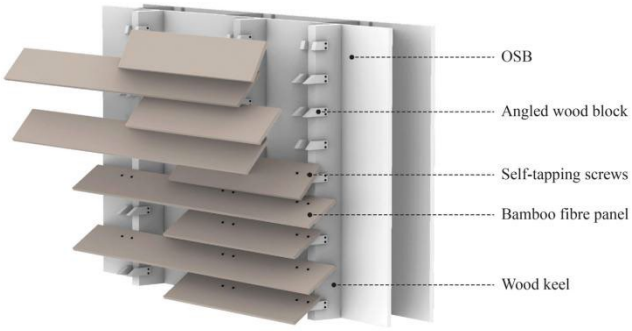


Figure 4. Assembly diagram of the non-standard building envelope

All of Aurora's components were pre-fabricated within a factory in Suzhou. The building was divided into 5 modules for the sake of road transportation. Before arriving on site in Zhang Jiakou, most of the building exterior parts have been installed except the bamboo panels and the mounting structure for solar system. It only took four hours to position and assemble the modules on-site (Figure 5).



Figure 5. From left to right: students cut bamboo panels; position a module to the foundation; and the on-site assembly of five modules

2.3. MR-AIDED CONSTRUCTION

To reduce on-site construction complexity, the implementation team adopted Fologram with HoloLens and handheld smart devices to assist the assembly of bamboo panels and mounting structures for the solar system. A portable hotspot device was brought to the site to ensure a live link between the holographic instructions and the parametric model. By setting parameters to control model movement, rotation, and display colours, implementers were able to fine-tune the alignment between virtual and actual objects, and to adjust projection clarity according to the lighting condition at the time. The MR implementation team relied on ArUco marker to map the design model to the as-built environment. Because both HoloLens and smart devices track 2D markers through a front camera, the team placed three 600*600mm ArUco markers to the designated positions. This allowed the implementers to register from a relatively further distance and to amend alignment at need.

In the construction of Aurora's non-standard envelope, the implementation team applied five different colours to identify flipping and non-flipping panels (Figure 6). Through HoloLens and iPad, this colourized model was holographically presented to the field of vision of the implementers. Also, to make it convenient for camera tracking, the implementers placed the ArUco markers both vertically on the wall and horizontally on the ground. A parametrically flexible projection helped to reduce the misalignment that has been caused by the deviation from previous module assembly and the issue of model drifting. In the beginning, a student implementer wearing HoloLens had labelled all the flipping types on each wood keel so that others without holographic instructions were able to install angled wood blocks simultaneously.

Since all the installation work was carried out manually by inexperienced students, construction deviation was inevitable. The augmented implementers, fortunately, could make ad-hoc adjustments based on the projection of an initial design idea, and make up construction errors to the greatest extent (Figure 7). To illustrate this point, Aurora's non-standard facade was finished within 3 days by 12 people.



Figure 6. Holographic projection of Aurora's facade on actual building

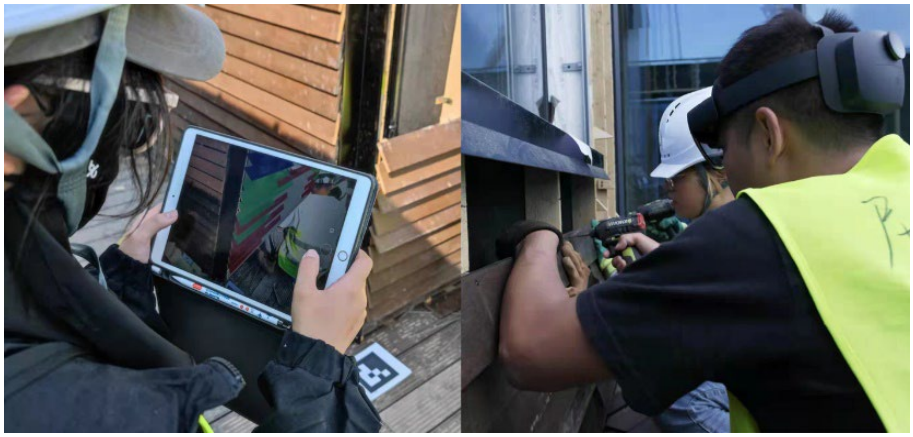


Figure 7. 3D registration with iPad (left); augmented panel installation with HoloLens (right)

The implementation team also adopted MR technology to assist the positioning and the assembly of rooftop mounting structure. Made of laminated bamboo strips, the mounting structure of solar system consists of 112 components that were pre-fabricated in Fujian Province. The plan to preassemble the structure on the ground and lift in one piece turned out to be impractical due to site restrictions and a potential irreversible deformation. The implementation team, instead, had decided to install these components one at a time directly on the roof, which extensively increased the construction difficulty. The holographic projection of the design model provided the

implementers a general sense of the structure in place (Figure 8) and enabled them to coordinate with the crane operator more effectively.

Since an accumulation of construction deviation is inevitable in this case, MR technology offered the implementer an approach to increase construction tolerance. By accordingly modifying the design model based on the structural components which have been installed on the roof, the implementers were able to continuously assess constructability and reshaped the uninstalled ones. For example, some of the structural components had to be trimmed in length and their pre-drilled holes were also re-made so that they could fit in place.



Figure 8. Holographic projection of the overall structure model

2.4. AS-BUILT ANALYSIS

Virtual-actual feedback is crucial to ensure implementation accuracy in a high-touch construction process. The above-mentioned materialisation processes can potentially lead to the actual building deviates greatly from the design model. Therefore, the project team had retrieved the as-built information and compared it with the initial design data, so that they were able to identify areas demand adjustment or reassembly.

Aurora's exterior as-built information was collected using a FARO laser scanner (Figure 9). The point cloud data was generated from in total seven scanning stations, then being integrated and converted to mesh objects. By overlaying the initial design model on top of the recreated one, the project team was able to calculate virtual-actual differences. Taking part of the south facade as an example, the project team appointed the cuboid geometry of the sunshade system as the overlaying reference, then analysed the differences between the scanned mesh model to the designed model. The green area in the figure10 means the deviation value is within 10mm. However, the red and blue dots represent the built panels are in front of or behind the designed ones. Deviations range from 10mm to 50mm and darker colours indicate larger distances. Based on the result, the implementation team adjusted some of the installed bamboo panels. Again, through Fologram and HoloLens, revising instructions were holographically presented to on-site implementers so that they were able to locate and fix the problem in short time.

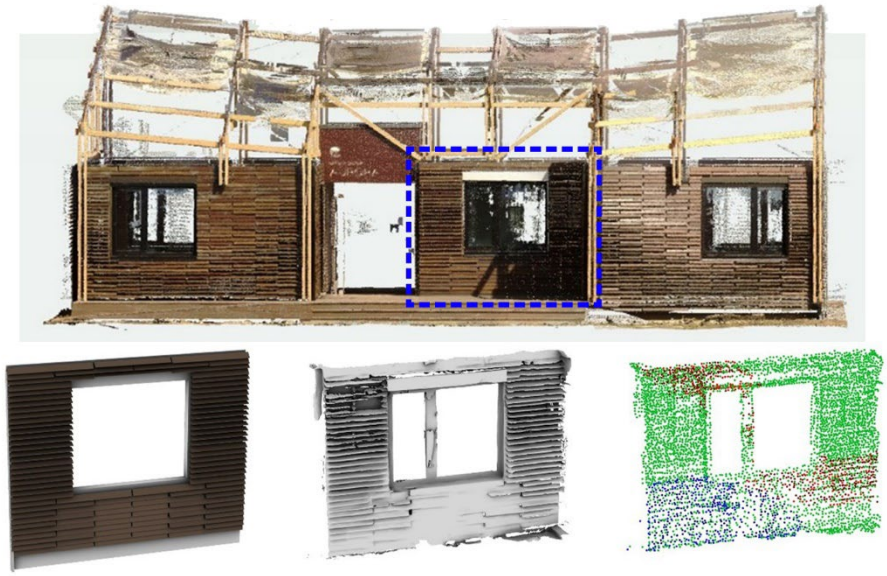


Figure 9. Aurora's exterior as-built information by laser scanner (up) and error analysis of a façade from the South elevation(down)

3. In conclusion

The use of MR technology in the construction of Aurora's non-standard envelop and the mounting structure for the solar system demonstrates how the participation of MR leads to the reform of the construction process, and reveals the potentially wide application of the highly effective low-tech implement. By removing the redundant annotation work from a design-to-build workflow, MR technology increases architects' agency in design exploration and materialisation. This study presents a real-time interaction between virtual and actual matters. Opposing to pre-programmed 'high-tech' robotic fabrications and assemblies, the MR-based human-machine collaboration promises architects an increasing role in today's post-digital discourses. Instead of chasing zero-tolerance implementation and avoiding any potential deviations, MR-aided design practices aim to increase construction allowance by injecting computation into human operations. This sheds light on the construction activities that still heavily rely on manual labour. Especially for non-standard building practices that are carried out within developing regions, the involvement of holographic instructions makes the entire implementation process more systemic and resilient to meet design expectations. Also, the easy access to hardware devices like HoloLens and smartphones, together with their MR applications such as Fologram and TwinBuild, lowers the threshold for adopting this technology in action.

Compared to small-scale architectural experiments or full-scale research projects that have taken place in controlled lab environments, the holographic construction of Aurora investigated the influence of outdoor matters, like sunlight and temperature, to a precise overlay of the digital information. The study discusses a marker-based

approach for on-site alignment. The use of parametric associative modelling allows the implementers to procedurally control data outputs. In return, this allows them to absorb process uncertainties and to accommodate project-specific idiosyncrasies. Demanding feedback from an as-built condition and followed by corresponding adjustments also indicate a potential workflow change in project delivery. This leads to a more productive dialogue between designers and implementers. With MR's accessibility and quick response, architects might delay decision-making and further expand the design solution space.

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